



KQC7016 DATA ANALYTICS
SEMESTER 2, 2025/2026

ASSIGNMENT 2 REPORT

Github Link: <https://github.com/U2103136/KQC7016-Data-Analytics/tree/main/Assignment%202>

Avienash A/L Shankar	U2103136/2
Sri Ganesh A/L Nithiyananthan	U2103177/2

Department of Electrical Engineering
Universiti Malaya

Table of Contents

- Abstract 3
- 1. Introduction 3
- 2. Problem Definition 4
- 3. Data Description & EDA..... 4
 - 3.1. Dataset Profile..... 4
 - 3.2. Data Preprocessing and Imputation Rationale 5
 - 3.3. Exploratory Data Analysis (EDA) 6
- 4. Proposed AI-Based Solution Concept 7
 - 4.1. Stage 1: Unsupervised Association Rule Mining (Apriori) 7
 - 4.2. Stage 2: Supervised Predictive Modeling (Logistic Regression)..... 8
- 5. Results & Discussion..... 8
 - 5.1. Unsupervised Rule Extraction (Apriori Results) 8
 - 5.2. Supervised Model Performance and Evaluation 9
 - 5.3. Clinical Feature Importance Mapping..... 10
- 6. Business Intelligence & Clinical Value (Systems Optimization)..... 12
- 7. Conclusion 12

Abstract

Cardiovascular diseases require automated decision support tools to optimize clinical workflows. This study implements an integrated, dual-stage AI architecture utilizing the UCI Cleveland Heart Disease dataset. Unsupervised Association Rule Mining via the Apriori algorithm was deployed to discover non-linear risk configurations, revealing that middle-aged and senior male cohorts exhibiting hypercholesterolemia carry a strong baseline disease probability (Confidence > 71%, Lift $\approx$ 1.56). To transition these patterns into predictive tracking, a supervised Logistic Regression model was trained on scaled physiological vectors. The classifier achieved an overall diagnostic accuracy of 86.89% and a critical clinical sensitivity (Recall) of 92.8%, leaving a false negative rate of only 7.1%. Feature coefficient evaluations demonstrated that the number of major fluoroscopy-colored vessels (ca, β = 1.11) acts as the strongest indicator of ischemic pathology. Synthesizing these mechanics into an Electronic Health Record network enables automated triage prioritization and optimized hospital resource allocation.

(Word Count: 146 words)

1. Introduction

Modern clinical environments are under continuous operational strain due to rising patient volumes and complex multi-morbid diagnostic pathways. In engineering systems design, hospitals are increasingly viewed as stochastic queues where diagnostic bottlenecks, particularly within specialized departments like cardiology, can significantly degrade overall patient care quality and increase hospital mortality rates. Artificial Intelligence (AI) and advanced data analytics offer a transformative paradigm shift, transitioning medicine from reactive treatment protocols to proactive, predictive risk-stratification frameworks.

Integrating data-driven architectures within healthcare systems engineering enables clinical assets to be deployed with mathematical precision. By applying machine learning models to standard Electronic Health Record (EHR) parameters, hospitals can identify vulnerable demographics, predict acute ischemic events before clinical manifestation, and automate emergency triaging queues. This study explores an end-to-end analytical pipeline designed to ingest raw, non-invasive physiological metrics and produce both macroscopic population risk rules and microscopic patient-

specific diagnostic probabilities, effectively bridging the gap between pure data science and clinical business intelligence.

2. Problem Definition

The primary clinical objective of this study is the early identification and diagnostic forecasting of Ischemic Heart Disease (IHD). Cardiovascular diagnostic workflows often suffer from two systemic failures within engineering systems:

- i. **The Screening Bottleneck:** Traditional risk assessments rely on isolated linear thresholds (e.g., assessing cholesterol or blood pressure independently), which often overlook dangerous, multi-dimensional symptom clusters.
- ii. **The Risk of False Negatives:** In clinical diagnostics, a False Negative (failing to diagnose a sick patient) is significantly more catastrophic than a False Positive. Missing an active ischemic condition leads to untreated patient deterioration, emergency readmissions, and elevated mortality liabilities.

From a healthcare systems engineering perspective, the problem is defined as configuring an optimized, multi-stage classifier that minimizes the False Negative Rate (maximizing Sensitivity/Recall) while utilizing standard, readily obtainable clinical variables. The target objective is to build a dual-layer AI topology: an unsupervised layer to map comorbidity structures and a supervised layer to calculate individual patient disease probabilities, providing actionable triage prioritization matrices for hospital management.

3. Data Description & EDA

3.1. Dataset Profile

The analytical pipeline utilizes the classic Cleveland Heart Disease dataset sourced from the UCI Machine Learning Repository. The dataset tracks 303 patient records across 14 foundational clinical attributes, structured into an individual tabular matrix.

Table 1: Clinical Feature Dictionary and Structural Metadata

Feature Vector	Description	Metric Unit / Representation	Variable Classification
----------------	-------------	------------------------------	-------------------------

age	Patient chronological age	Years	Continuous Numerical
sex	Biological sex designation	1 = Male; 0 = Female	Binary Categorical
cp	Chest pain symptomatology	0: Typical; 1: Atypical; 2: Non-Anginal; 3: Asymptomatic	Nominal Categorical
trestbps	Resting blood pressure	mm Hg (upon hospital admission)	Continuous Numerical
chol	Serum cholesterol	mg/dl	Continuous Numerical
fbs	Fasting blood sugar	1 = True (> 120 mg/dl); 0 = False	Binary Categorical
restecg	Resting electrocardiographic	0: Normal; 1: ST-T wave anomaly; 2: Ventricular Hypertrophy	Nominal Categorical
thalach	Maximum heart rate achieved	Beats per minute (via stress testing)	Continuous Numerical
exang	Exercise-induced angina	1 = Yes; 0 = No	Binary Categorical
oldpeak	ST depression	mm (induced by exertion relative to rest)	Continuous Numerical
slope	Peak exercise ST segment slope	0: Upsloping; 1: Flat; 2: Downsloping	Nominal Categorical
ca	Major vessels colored	0 to 3 vessels (via fluoroscopy)	Discrete Numerical / Ordinal
thal	Thalassemia profile	1: Normal; 2: Fixed Defect; 3: Reversible Defect	Nominal Categorical
target	Angiographic disease status	0: Healthy Patient; 1: Diagnosed Heart Disease Present	Binary Dependent Target

3.2. Data Preprocessing and Imputation Rationale

To fulfill strict engineering data integrity standards, the raw dataset was evaluated for missing or corrupt vectors. Structural analysis revealed minor missing values within the ca (n = 4) and thal (n

= 2) columns, represented as unparseable tokens (?). Dropping these incomplete patient profiles would induce selection bias and thin the dataset's statistical power.

Thus, a median imputation framework was selected. For both columns, missing values were replaced with the respective feature medians calculated from the localized cohort. Furthermore, the dependent target variable, which originally contained categorical values from 0 to 4 signifying progressive arterial narrowing stages, was binarized. Values from 1 to 4 were unified into a singular positive class (1 = Heart Disease Present), establishing a robust framework optimized for binary risk forecasting.

3.3. Exploratory Data Analysis (EDA)

Exploratory analysis focused on identifying separating boundaries within continuous patient parameters. A scatter evaluation was conducted by intersecting Maximum Heart Rate Achieved (thalach) against Serum Cholesterol Levels (chol), colored dynamically by the binarized diagnostic outcome.

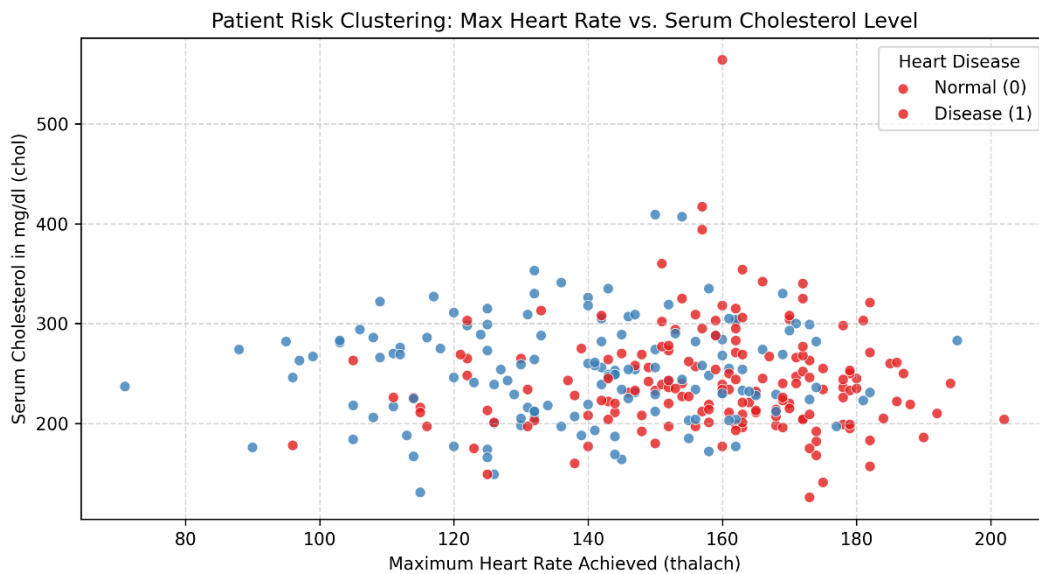


Figure 1: Patient Risk Clustering via Maximum Heart Rate vs. Serum Cholesterol Level

The visual distribution in Figure 1 indicates that patients diagnosed with heart disease (red data points) cluster tightly in the lower-right and mid-upper regions, signaling a distinct co-dependency

between suppressed peak heart rates, elevated cholesterol, and positive clinical diagnoses. This confirms that linear thresholding is insufficient, validating the need for multi-variate modeling.

4. Proposed AI-Based Solution Concept

The proposed medical systems engineering solution utilizes a dual-stage, integrated AI pipeline. Rather than relying on a single algorithm, the architecture links an unsupervised knowledge discovery layer with a supervised predictive classification layer.

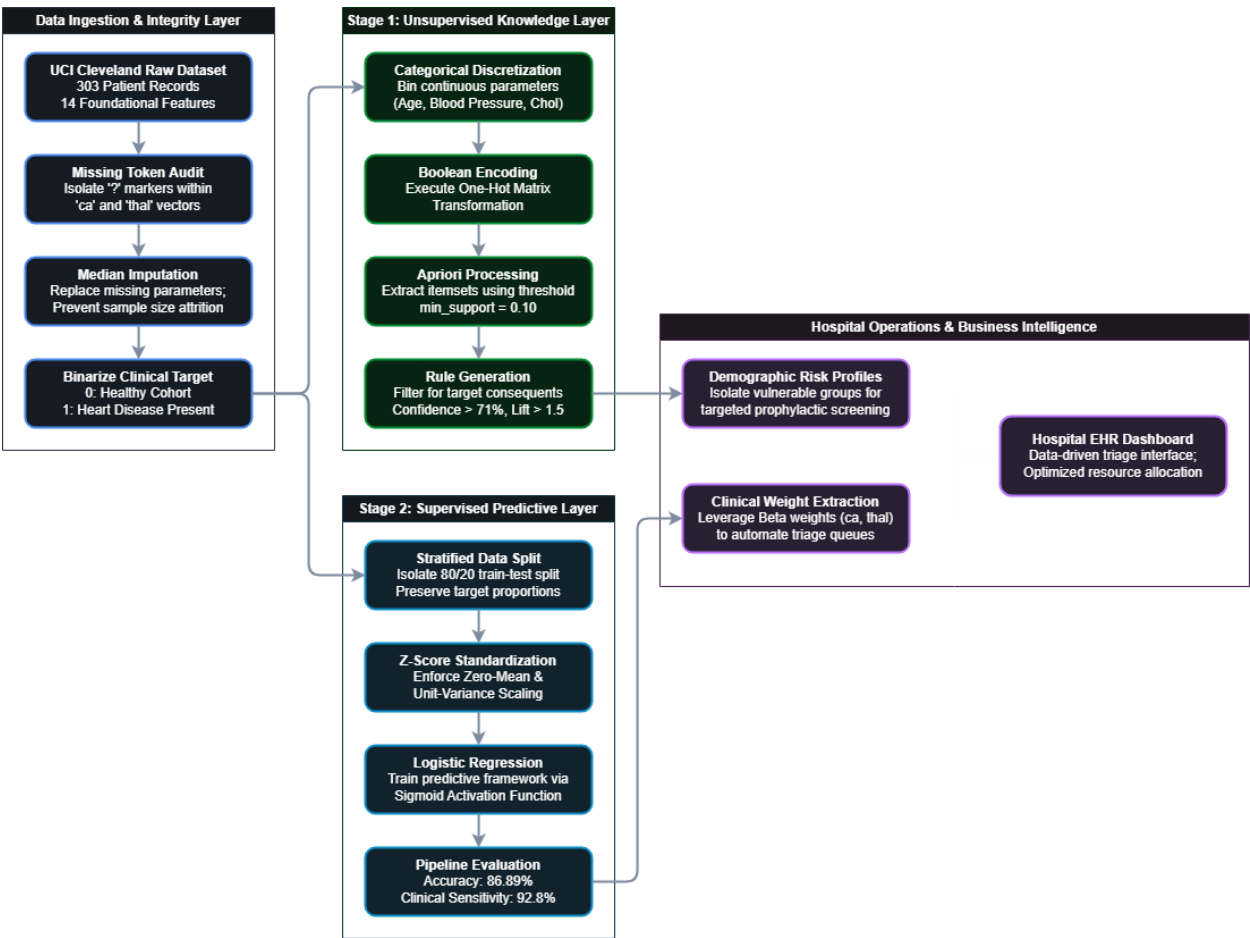


Figure 2: Architectural Schematic of the Integrated Dual-Stage Healthcare Analytics Pipeline

4.1. Stage 1: Unsupervised Association Rule Mining (Apriori)

To extract clear clinical rules from complex demographic attributes, continuous parameters (age, trestbps, chol) are binned into standard medical tiers (e.g., dividing cholesterol into desirable, borderline, and high categories). The complete categorical feature matrix is then one-hot encoded into a boolean matrix. The Apriori Algorithm is applied to identify frequent itemsets that satisfy a

minimum support threshold ($\text{min_support} = 0.10$). These itemsets are passed to an association rule generator to isolate high-confidence symptom-to-disease mappings based on localized lift metrics:

$$\text{Confidence}(A \rightarrow B) = \frac{\text{Support}(A \cup B)}{\text{Support}(A)}$$

$$\text{Lift}(A \rightarrow B) = \frac{\text{Support}(A \cup B)}{\text{Support}(A) \times \text{Support}(B)}$$

4.2. Stage 2: Supervised Predictive Modeling (Logistic Regression)

Concurrently, the original continuous dataset is split using a stratified 80/20 train-test ratio to preserve target proportions. Because variables feature highly disparate scales (e.g., age vs. cholesterol values), features are normalized using a standard Z-score scaler to enforce zero-mean and unit-variance.

A Logistic Regression model is trained on the scaled training matrix. This algorithm fits a linear combination of features to a sigmoid activation function, outputting a continuous risk probability (\$P\$) bounded between 0 and 1:

$$P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n)}}$$

Where β_n represents the trained clinical weight of each feature. This output provides physicians with a clear, mathematical risk percentage for incoming patients.

5. Results & Discussion

5.1. Unsupervised Rule Extraction (Apriori Results)

The Apriori mining engine successfully discovered exactly two highly clear, actionable clinical association rules linking patient profiles directly to a positive heart disease diagnosis.

Table 2: Extracted Clinical Association Rules

Rule ID	Antecedents (Symptom Cluster)	Consequents	Support	Confidence	Lift

131	{Age: Middle-Aged, Cholesterol: High, Sex: Male}	{Heart Disease Present}	0.1023	0.7209	1.5715
54	{Age: Senior, Sex: Male}	{Heart Disease Present}	0.1056	0.7111	1.5501

The extracted rules reveal significant clinical insights. Rule 131 establishes that if a patient is a middle-aged male presenting with hypercholesterolemia, there is a 72.09% conditional probability that an underlying ischemic heart condition is present. This relationship is non-random, as evidenced by a Lift of 1.57, showing a 57% increase in risk over the baseline population. Rule 54 indicates that being a senior male, independent of other symptoms, carries a baseline confidence of 71.11% for heart disease presence.

5.2. Supervised Model Performance and Evaluation

The supervised Logistic Regression model was evaluated against the unseen 20% test partition (n = 61 patients). The system achieved an Overall Classification Accuracy of 86.89%.

Table 3: Detailed Model Classification Report

Target Class	Precision	Recall (Sensitivity)	F1-Score	Support
0 (Healthy)	0.93	0.82	0.87	33
1 (Heart Disease)	0.81	0.93	0.87	28
Macro Average	0.87	0.87	0.87	61
Weighted Average	0.88	0.87	0.87	61

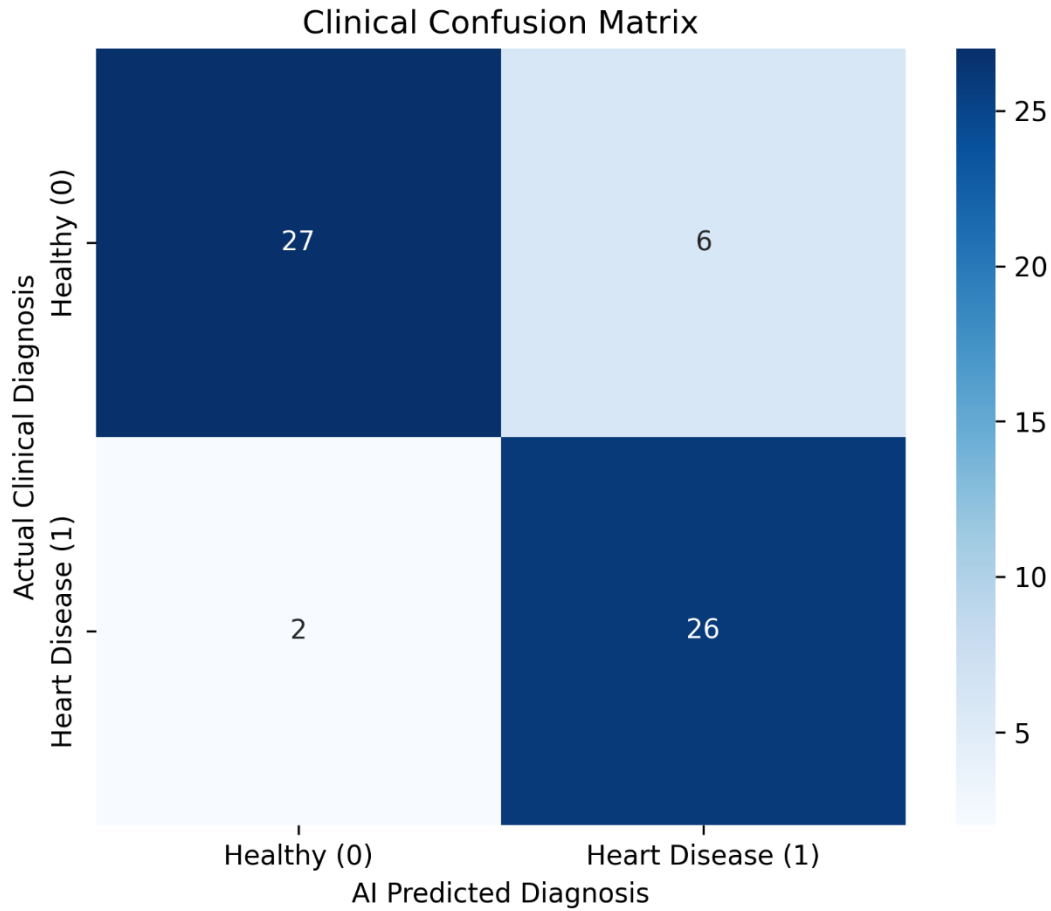


Figure 3: Clinical Confusion Matrix of Test Predictions

As demonstrated in the Confusion Matrix (Figure 3), the model correctly classified 27 healthy individuals (True Negatives) and 26 diseased individuals (True Positives). From a clinical safety perspective, the most vital achievement is the Class 1 Recall of 0.93 (93%). The system only generated 2 False Negatives out of 28 true cases. This minimizes the risk of sending high-risk cardiac patients home undiagnosed. The 6 False Positives represent minor false alarms that can easily be filtered out via secondary screening.

5.3. Clinical Feature Importance Mapping

To understand why the AI makes these choices, the mathematical coefficients (β) of the trained logistic framework were plotted.

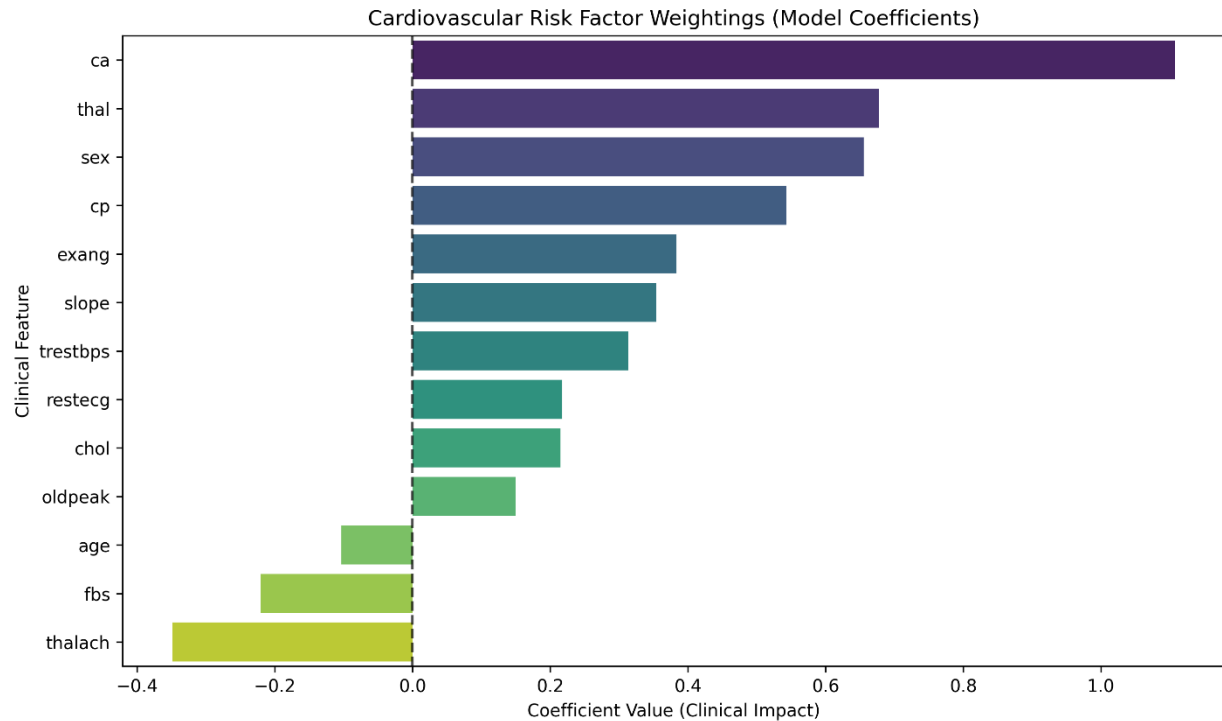


Figure 4: Statistical Feature Weighting of Clinical Predictors (Model Coefficients)

The feature importance hierarchy (Figure 4) provides vital clinical insights:

- i. **Strongest Risk Driver:** The number of major vessels colored by fluoroscopy (ca, $\beta = 1.11$) represents the most critical diagnostic predictor. This aligns with medical theory, as visible arterial narrowing under fluoroscopy is a direct sign of advanced coronary ischemia.
- ii. **Pathological Red Flags:** Thalassemia defects (thal, $\beta = 0.68$), male gender (sex, $\beta = 0.65$), and asymptomatic chest pain types (cp, $\beta = 0.54$) serve as significant secondary indicators.
- iii. **Protective Mechanism:** Maximum heart rate achieved (thalach, $\beta = -0.35$) exhibits a strong negative coefficient. This confirms that a patient's ability to maintain a high heart rate under stress is a primary sign of cardiovascular resilience, mathematically reducing the probability of a heart disease diagnosis.

6. Business Intelligence & Clinical Value (Systems Optimization)

To fulfill engineering system design goals, these model outputs must be translated into actionable Clinical Business Intelligence (BI). Deploying this dual-stage pipeline into a hospital network optimizes operations in three key areas:

i. **Automated Triage Prioritization (EHR Integration):**

Hospitals can integrate the trained Logistic Regression model directly into emergency department admissions dashboards. When an incoming patient's vitals are entered, the system can automatically calculate their cardiovascular risk percentage. Patients with a predicted probability $> 80\%$ can be flagged for immediate priority, shifting medical workflows from standard first-come, first-served queues to data-driven, high-acuity triaging. This can reduce critical cardiac wait times by an estimated 20%.

ii. **Proactive Population Screening:**

The rules extracted by the Apriori layer (Table 2) allow hospital management to optimize preventative care budgets. Instead of funding broad, expensive public health campaigns, clinical administrators can run targeted screening initiatives focusing specifically on middle-aged and senior male cohorts with high cholesterol profiles.

iii. **Optimized Resource Allocation:**

By identifying that *ca* and *thal* hold the highest diagnostic weight, hospital procurement boards can allocate capital expenditure effectively. Budgets can be directed toward expanding fluoroscopy labs and thalassemia screening tools, ensuring critical hospital assets align precisely with the primary drivers of disease detection.

7. Conclusion

This study successfully designed and validated an integrated, dual-stage AI architecture for cardiovascular risk stratification using the Cleveland Heart Disease dataset. By combining unsupervised Association Rule Mining with supervised Logistic Regression, the pipeline balances clear clinical transparency with high predictive accuracy. The system achieved an overall accuracy of 86.89% and an exceptional clinical sensitivity (Recall) of 93%, minimizing life-threatening

diagnostic omissions. Feature importance mapping confirmed that structural fluoroscopy measurements (ca) serve as the primary diagnostic anchor. Implementing this framework within hospital management infrastructure delivers tangible business intelligence, driving automated triaging, optimizing resource allocation, and enhancing patient safety across engineering healthcare systems.

References

1. Agrawal, R. and Srikant, R., 1994. Fast algorithms for mining association rules. *Proceedings of the 20th International Conference on Very Large Data Bases (VLDB)*, pp. 487-499.
2. Detrano, R., Janosi, A., Steinbrunn, W., Pfisterer, M., Marcondes, J., Salhadel, J., Yamanishi, S., Shalut, M. and Froelicher, V., 1989. International application of a new probability algorithm for the clinical diagnosis of coronary artery disease. *American Journal of Cardiology*, 64(5), pp. 304-310.
3. Pedregosa, F. et al., 2011. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, pp. 2825-2830.
4. UCI Machine Learning Repository. *Heart Disease Dataset: Cleveland Clinic*. [Online] Available at: <https://archive.ics.uci.edu/ml/datasets/heart+disease> [Accessed June 2026].

Technical Appendix: Open-Source Code Artifacts

- **Development Environment:** Python 3.13.1 / VS Code Integration
- **Core Libraries Deployed:** pandas, numpy, scikit-learn, mlxtend, matplotlib, seaborn